

OS PRACTICAL

PRACTICAL1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

OUTPUT:

```
Rashmika@LAPTOP-UNQ0CN  x + v - □ ×
Rashmika@LAPTOP-UNQ0CNLA:~$ mkdir practical1
Rashmika@LAPTOP-UNQ0CNLA:~$ cd practical1
Rashmika@LAPTOP-UNQ0CNLA:~/practical1$ nano practical1.c
Rashmika@LAPTOP-UNQ0CNLA:~/practical1$ gcc practical1.c -o practical1
Rashmika@LAPTOP-UNQ0CNLA:~/practical1$ ./practical1
Enter Linux command: uname

Parent Process
Parent PID : 577

Child Process
Child PID : 592
Linux
Child process completed.
Rashmika@LAPTOP-UNQ0CNLA:~/practical1$ ./practical1
Enter Linux command: lscpu

Parent Process
Parent PID : 604

Child Process
Child PID : 605
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-1355U
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               5222.39
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                        r pge mca cmov pat pse36 clflush mmx fxsr sse s
                        se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                        _tsc rep_good nopl xtopology tsc_reliable nonst
```

```
_tsc rep_good nopl xtopology tsc_reliable nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
opcmt tsc_deadline_timer aes xsave avx f16c rdr
and hypervisor lahf_lm abm 3dnowprefetch ssbd i
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrm md_clear serialize ibt flu
sh_lld arch_capabilities
```

Virtualization features:

```
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
```

Caches (sum of all):

```
L1d: 288 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
```

NUMA:

```
NUMA node(s): 1
NUMA node0 CPU(s): 0-11
```

Vulnerabilities:

```
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled v
ia prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user
pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB con
ditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_
S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
```

Child process completed.

```
029 pts/0    00:00:00 ps
Rashmika@LAPTOP-UNQ0CNLA:~/practical1$ ./practical1
Enter Linux command: ps
```

```
Parent Process
Parent PID : 639
```

```
Child Process
Child PID : 640
  PID TTY          TIME CMD
  312 pts/0    00:00:00 bash
  639 pts/0    00:00:00 practical1
  640 pts/0    00:00:00 ps
Child process completed.
```

```
Rashmika@LAPTOP-UNQ0CNLA:~/practical1$ ./practical1
Enter Linux command: top
```

```
Parent Process
Child Process
Parent PID : 650
Child PID : 651
top - 05:48:42 up 1:02, 1 user, load average: 0.01, 0.02, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7762.6 total, 7148.5 free, 551.9 used, 213.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7210.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24348	15416	11564	S	0.0	0.2	0:00.65	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50420	16788	15508	S	0.0	0.2	0:00.19	systemd-journal
78	systemd+	20	0	22416	14496	12008	S	0.0	0.2	0:00.07	systemd-resolve
85	root	20	0	35236	12340	9260	S	0.0	0.2	0:00.31	systemd-udev
107	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.04	chronyd-starter
108	root	20	0	4472	3136	2888	S	0.0	0.0	0:00.01	cron
112	message+	20	0	8816	5408	4728	S	0.0	0.1	0:00.05	dbus-daemon
155	root	20	0	44096	29872	14732	S	0.0	0.4	0:00.22	networkd-dispat
163	root	20	0	18436	9412	8168	S	0.0	0.1	0:00.08	systemd-logind
179	syslog	20	0	220548	5880	4584	S	0.0	0.1	0:00.06	rsyslogd
200	root	20	0	5492	2764	2580	S	0.0	0.0	0:00.00	agetty
227	_chrony	20	0	23936	11480	7244	S	0.0	0.1	0:00.11	chronyd
238	_chrony	20	0	12096	2468	1824	S	0.0	0.0	0:00.01	chronyd
266	root	20	0	123460	32548	17936	S	0.0	0.4	0:00.18	unattended-upgr
309	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
310	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.09	Relay(312)
312	Rashmika	20	0	6268	5600	3928	S	0.0	0.1	0:00.07	bash
314	root	20	0	8644	5612	4772	S	0.0	0.1	0:00.01	login
372	Rashmika	20	0	22516	13424	10964	S	0.0	0.2	0:00.09	systemd
376	Rashmika	20	0	22916	4092	2088	S	0.0	0.1	0:00.00	(sd-pam)
402	Rashmika	20	0	6136	5404	3856	S	0.0	0.1	0:00.01	bash
650	Rashmika	20	0	2768	1880	1784	S	0.0	0.0	0:00.00	practical1
651	Rashmika	20	0	8440	6220	3896	R	0.0	0.1	0:00.95	top

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Rashmika@LAPTOP-UNQ0CNLA:~/practical1\$./practical1

Enter Linux command: lsblk

Parent Process

Parent PID : 618

Child Process

Child PID : 619

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	356.9M	1	disk	
sdb	8:16	0	159.4M	1	disk	
sdc	8:32	0	2G	0	disk	[SWAP]
sdd	8:48	0	1T	0	disk	/mnt/wslg/distro /

Child process completed.